



Department of Computer Science
Faculty of Computing and Information Technology (FCIT)
University of the Punjab

HelmX

An AI Powered Smart Helmet

Final Year Project Proposal

Session 2025-2026

A project submitted in partial
fulfilment of the BS in Computer
Science
Fall 2025

09 September 2025

Project Registration

Project ID (for office use)						
Nature of project	☐ Development ☒ R&D					
Area of specialization	Computer Vision, Web Development, Cloud, Android App Development, AI and ML, IOT, Software and Hardware Integration					
Project Group Members						
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Date: September 16, 2025

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Name of Supervisor: Ms. Tayyaba Tariq

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Project Abstract

Motorcycle accidents remain an important public health concern around the world, and Pakistan reports more than 10,000 deaths annually due to road accidents, which often involve two-wheeled vehicles. The HelmX project aims to address this pressing challenge by developing an AI-backed system for motorcycles that aims to improve rider safety, navigation, and the riding experience with real-time contextual observational monitoring and intelligent feedback. The system will be comprised of seven purposefully constructed main features: **(1)** Crash Detection utilizing sensors embedded within the device such as accelerometers and gyroscopes to determine impacts, automatically notifying the emergency services via GPS; **(2)** Drowsiness Detection using computer vision algorithms on front-facing camera images and videos of the rider identifying both eye closure and head pose, which will generate both audio and visual alerts; **(3)** Smart Navigation which optimizes real-time recommendations and route suggestions by avoiding traffic and incorporating weather recommendations; **(4)** Capture images or videos with the Dual Cameras (front and rear) of the motorcycle that can constantly record veterinary observable evidence, or at the request of the rider via voice command to provide images of the environment; **(5)** Voice Control and Entertainment Systems capable of performing online and offline commands for media playback, calls, and system controls, adaptable to ambient sound conditions; **(6)** Safety and Environment Monitoring System measuring air quality, temperature, and allergens; they will ultimately generate ride recommendations; **(7)** Complete Web and android application for ride analytics.

Optional features, such as gesture-based controls and posture correction, fuel estimation, and AI safety tutorials for maintenance, are expected to be prioritized based on the projected timelines for development. Elements that are not addressed in the context of the project include large-scale manufacturing of hardware, medical validation of the health metrics, and advanced hardware like dynamic visor tint.

HelmX employs IoT sensors, AI algorithms, and user-centric interfaces, with the aim of preventing accidents through alerts and information-driven conclusions to reduce risk by a considerable percentage. Rather than purely an academic initiative, this project will have implications for sustainable transport systems of urban mobility in developing nations.

Researchers will measure the results of the project based on the prototype's functionality, accuracy of user testing (>85% for detection), and supervisor/client agreement. The research aims to enable riders the technology to prioritize safety while not detracting from the positively perceived thrill of riding.

1. Introduction

Motorcycle riders are among the most vulnerable users on the road, accounting for a disproportionately high number of global traffic fatalities. The primary challenges include limited visibility, rider fatigue, and the cognitive overload of navigating while operating a vehicle. Existing solutions, such as standard helmets and basic Bluetooth devices, offer fragmented features but lack an integrated, intelligent system capable of proactive accident prevention and post-incident response.

Helm X is a comprehensive solution for addressing safety challenges of motorcycle riders. This project will design and develop a smart helmet that employs the use of Artificial Intelligence and the internet of things to establish a smarter, intuitive riding ecosystem. The specific goals of the project are to: **1)** Create a robust hardware prototype with embedded cameras, inertial measurement units (IMU), a battery and other related things; **2)** Implement the core AI algorithms to detect drowsiness, analyze accidents, and utilize computer vision **3)** Establish secure cloud infrastructure for storing data and alerting in real-time; and **4)** Create web and mobile interfaces for visualizing data and enabling controls of the systems. Success will be measured by the accurate functioning of the core AI models, the seamless integration of all hardware and software components, and positive feedback from user acceptance testing.

2. Success Criterion

The following verifiable criteria will outline the success of the Helm X initiative:

1. **Functional Prototype:** A working prototype of the helmet will be delivered with the successful implementation of at least the seven required core features (Crash Detection, Drowsiness Detection, Navigation, Dual Cameras, Voice Control and Entertainment System, Safety System and Environment Monitoring System).
2. **AI Accuracy:** The accuracy and recall rate of the drowsiness detection system and crash detection system must be >90% at least in controlled testing environments and the crash detection algorithm must achieve a false positive rate of <10%.
3. **System Integration:** The Android application and web portal will successfully communicate with the prototype helmet displaying real-time data analytics, historical analytics, and system configuration.
4. **User Acceptance:** The system must receive a positive score (e.g., >70 on the System Usability Scale) from a pilot group of at least 20 test riders.
5. **Documentation:** Final submission must include complete technical documentation and a research paper on the system's architecture and treatment of AI.

3. Related work

Research into motorcycle safety systems has thus far identified the promise of using AI, but does demonstrate opportunities for integrated, cost-effective solutions, thanks to further collaboration within the developing context.

- In 2025, Kim et al. created an AI-based smart motorcycle helmet that provided drowsiness detection through EEG signals and facial recognition detection with a 92% accuracy, but was limited to helmet use without a navigation option [1].
- A 2023 Sentiance implementation demonstrated crash detection features using a mobile device's sensors to provide alerts about a crash in real time. Their approach appealed to privacy preserving ML while lacking good modeling and safety system integrations in their approach [2].
- A 2025 study conducted by Chigee, detailing an AIO-6 system to enhance CarPlay AI, evaluated improved routing to destination through rider behavior, but their AI based system did not provide rider ergonomics like posture monitoring systems [3].
- The 2024 Rexing MTC1 study provides dual camera angle recording with a 1080p definition and allows for evidence collection, but it relies on perceptive video evidence without AI or event tagging implementations [4].
- The 2024 study on voice and different entertainment systems highlighted the advantages with Bluetooth controls and heads-up displays, however, adaptive audio systems based on the traffic flow and noise level currently remain unexplored [5].
- The IoT-based crash alert system explored by Sharma et al. (2023) applies ML systems to high-definition data compilations, while adding systems based on our work with sensor fusion in designing a motorcycle safety system [6].
- The concept of AR for navigation in helmets has been pioneered by companies like BMW, iC and Skully, but their solutions are prohibitively expensive and not tailored for the broader market [7].

Overall, there are features that meet the academic standards, however, we have yet to create a comprehensive system that integrates core safety features with the added details, for riders from low-resource communities. HelmX will go a step further, combining these disparate features into a single, integrated, and cost-effective solution. Unlike commercially available products, HelmX will emphasize open, modular design and utilizes the latest deep learning models to enhance accuracy in real-world settings. Additionally, the development of a full-stack analytical platform provides a value proposition not seen with available products or prototypes.

4. Project Rationale

HelmX evolved out of an urgency to apply technology to save lives in motorcycle accidents. Motorcycles are 28 times more likely to result in fatalities than cars because motorcyclists are often immobilized or unconscious before they can be evaluated and treated. The initiative is relevant because it will address this gap by providing instant automated alerts. The project is also intended to reduce the cognitive load on riders and provide situational awareness while navigating and performing other tasks in order to make riding less distracting while riding safer.

Ultimately, we are excited to execute a project that allows us to apply our expertise and education related to AI, cloud and software engineering to a contemporary issue that has social impact and social significance. We hope to learn about embedded systems integration, the challenges related to deploying AI in resource-constrained settings, and full stack product development processes specifically for ridesharing.

4.1 Aims and Objectives

The overarching aim is to create an intelligent helmet ecosystem that proactively prevents accidents and ensures rapid response when they occur as well as provide entertainment and environmental awareness to rider.

Primary Objectives:

- To design and assemble a helmet prototype with integrated sensors (IMU, cameras, microphone).
- Integrate dual cameras for evidence capture and voice-commanded media.
- To develop and train AI models for drowsiness and crash detection.
- To implement a real-time navigation system, voice control system and safety system.
- To build a secure cloud backend for data processing and alerts.
- To integrate a personalized AI peer for communication and entertainment during long tours
- Establish environment monitoring for air quality, temperature, weather which then provides related suggestions.
- To develop an Android app and a responsive web dashboard.

4.2 Scope of the Project

The project's scope includes the end-to-end development of the HelmX system. Deliverables include the physical prototype, all source code for the firmware, AI models, mobile app, and web dashboard, along with full documentation.

Inclusion:

In-scope features are the seven core functionalities:

- Smart crash detection system to indicate sudden movements or abnormal movements indicative of crash and automatically triggers alerts to predefined contacts with location data upon crash detection.
- Drowsiness Detection system using camera that detects drowsiness and fatigue and issues real time alerts.
- Integrates GPS and AI-based navigation for real-time route guidance and suggestions.
- Features dual cameras and IMUs for comprehensive environmental awareness. Supports real-time video recording, live streaming, or image capture for safety documentation and situational analysis.
- Enables hands-free operation through voice recognition for controlling navigation, communication, or media playback. Includes an integrated entertainment system for audio streaming (e.g., music, podcasts) via Bluetooth-connected devices. It may also include a real time AI peer as a ride companion.
- For safety and security, it will include proper authentication and authorization and will generate real time alerts if theft is detected.
- Complete web application and android app application for real time analytics and control.

Optional:

Several "Optional" features can also be included if time permits. These include:

- Gesture Control
- Posture Detection and suggestion
- Smart Sound Controlling according to traffic
- Distance Monitoring and Fuel Estimation system
- Smart Rider Feedback system
- Smart Public Awareness System of dangerous areas on road
- Solar Assisted Charging system
- Heads up Display with Augmented Reality
- Smart Speed Optimization System for constant speed with less brakes
- AI bike scanner and Tutorials provider to fix issues
- Smart Personalized Riding Coach

Exclusion:

Out-of-scope items include:

- Hardware manufacturing at scale
- Advanced hardware like dynamic tinting
- Medical-grade validation of health metrics
- Smart mesh network system with nearby riders
- Smart gamified safe ride score
- Smart battery optimization
- Research on effect of electromagnetic field from sensors and microcontrollers on human health

- Hardware lifetime due to environmental issues and privacy concerns and international standards.

5. Proposed Methodology and Architecture

The project will follow an Agile development methodology with 2–4 week sprints. The system architecture is divided into four layers:

- **Hardware Layer:** In first phase complete hardware required will be collected. The helmet unit equipped with cameras, IMU, GPS module, battery, and relevant things will be collected.
- **Edge AI Layer (Optional):** On-device processing for real-time tasks (drowsiness detection, crash detection) using TensorFlow Lite models to minimize latency and cloud dependency.
- **Cloud Layer:** A secure AWS IoT Core and Node.js backend to handle data ingestion, storage (MongoDB), and send SMS/email alerts via Twilio API in case of a crash.
- **Application Layer:** A React.js web dashboard and a Kotlin-based Android app for users to view rides, analytics, and configure settings.

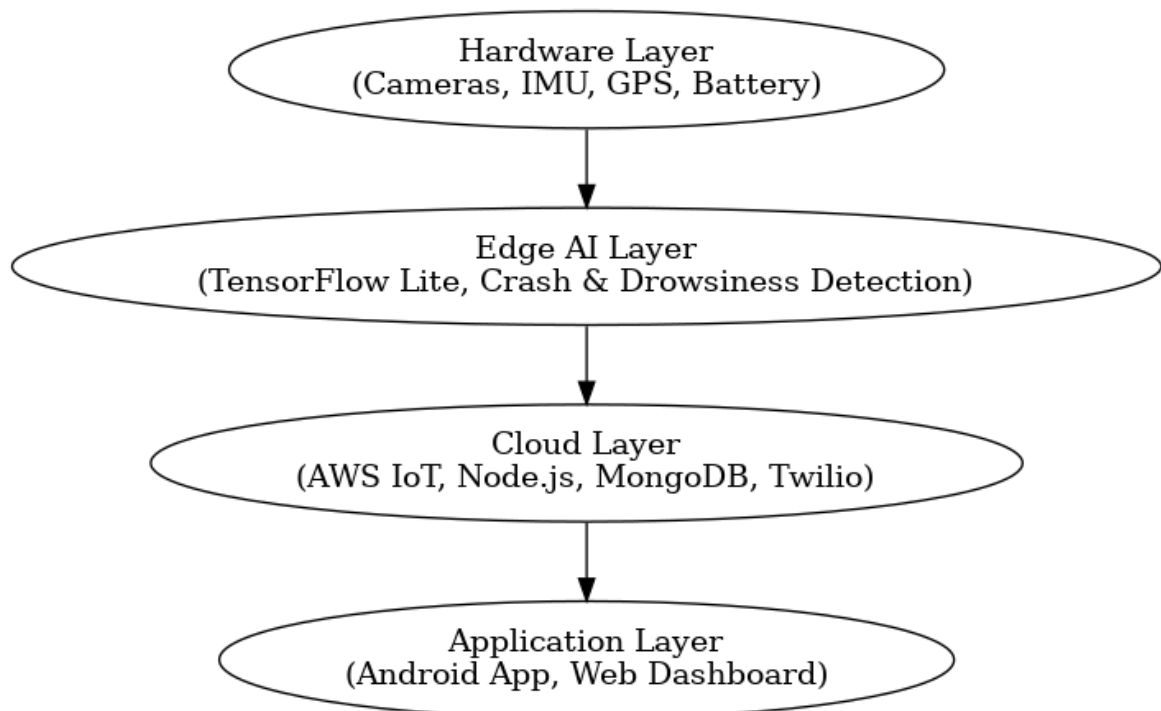


Fig 5.1 Flowchart of System Architecture

5.1 Algorithms:

A step-by-step process for a core feature, like crash detection, is:

- (i) IMU data is continuously read
- (ii) Data is fed into a pre-trained classifier on the edge device
- (iii) If a crash is predicted, the device triggers an alert
- (iv) GPS coordinates and crash data are sent to the cloud
- (v) The cloud service notifies emergency contacts and logs the event.

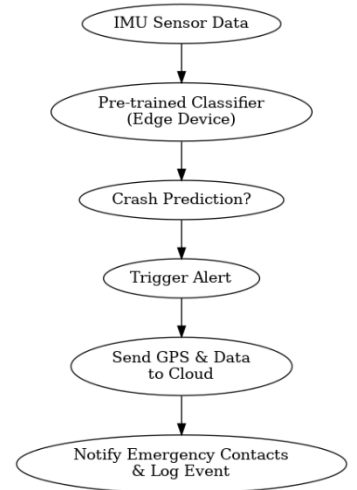


Fig 5.2 Crash Detection Flowchart

Similarly for drowsiness detection:

- (i) The process begins by receiving a constant video feed from the front-facing camera.
- (ii) The video feed is processed by a computer vision model, likely using libraries such as OpenCV and a Convolutional Neural Network (CNN).
- (iii) The model extracts specific features from the video, such as eye closure and head pose.
- (iv) The extracted features are analyzed to determine if the rider is drowsy.
- (v) If drowsiness is detected, the system generates both audio and visual alerts to notify the rider.

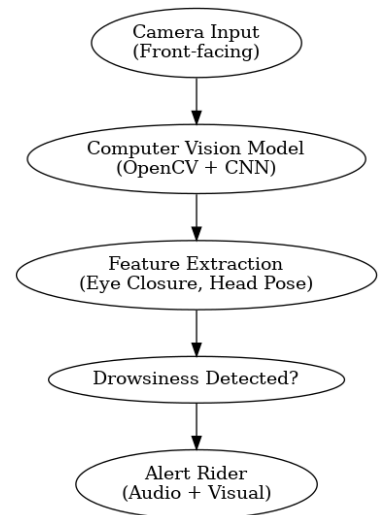


Fig 5.3 Drowsiness Detection Flowchart

Furthermore, for smart route suggestion:

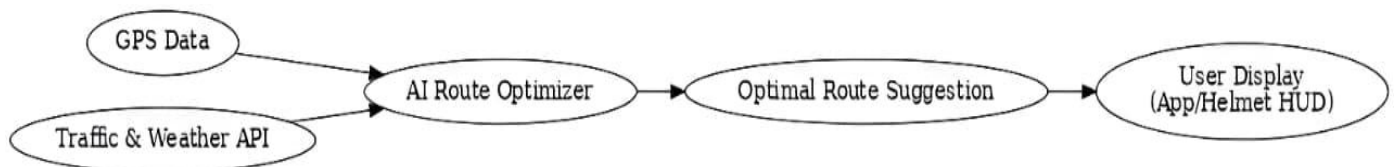


Fig 5.4 Smart Route Suggestion Flowchart

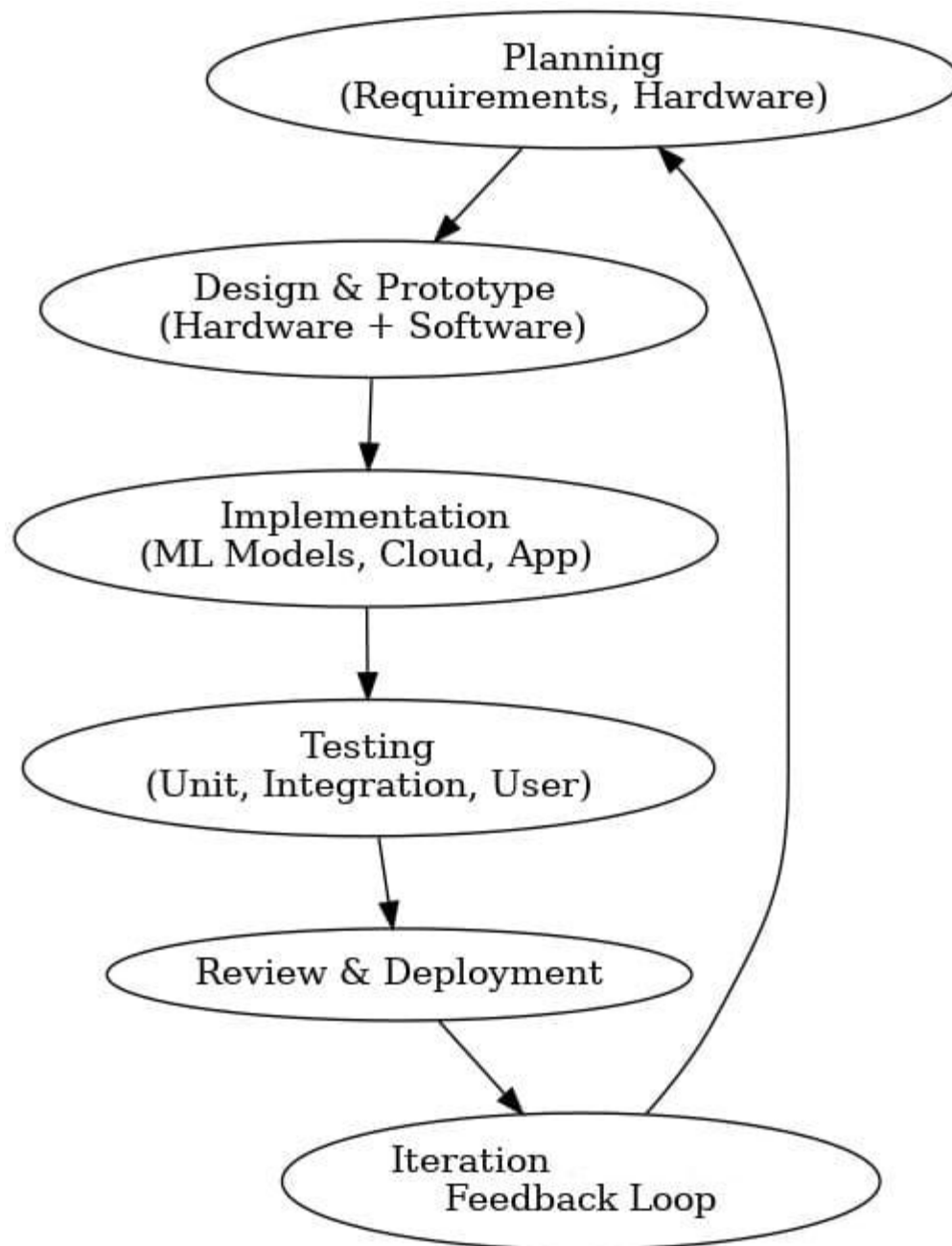
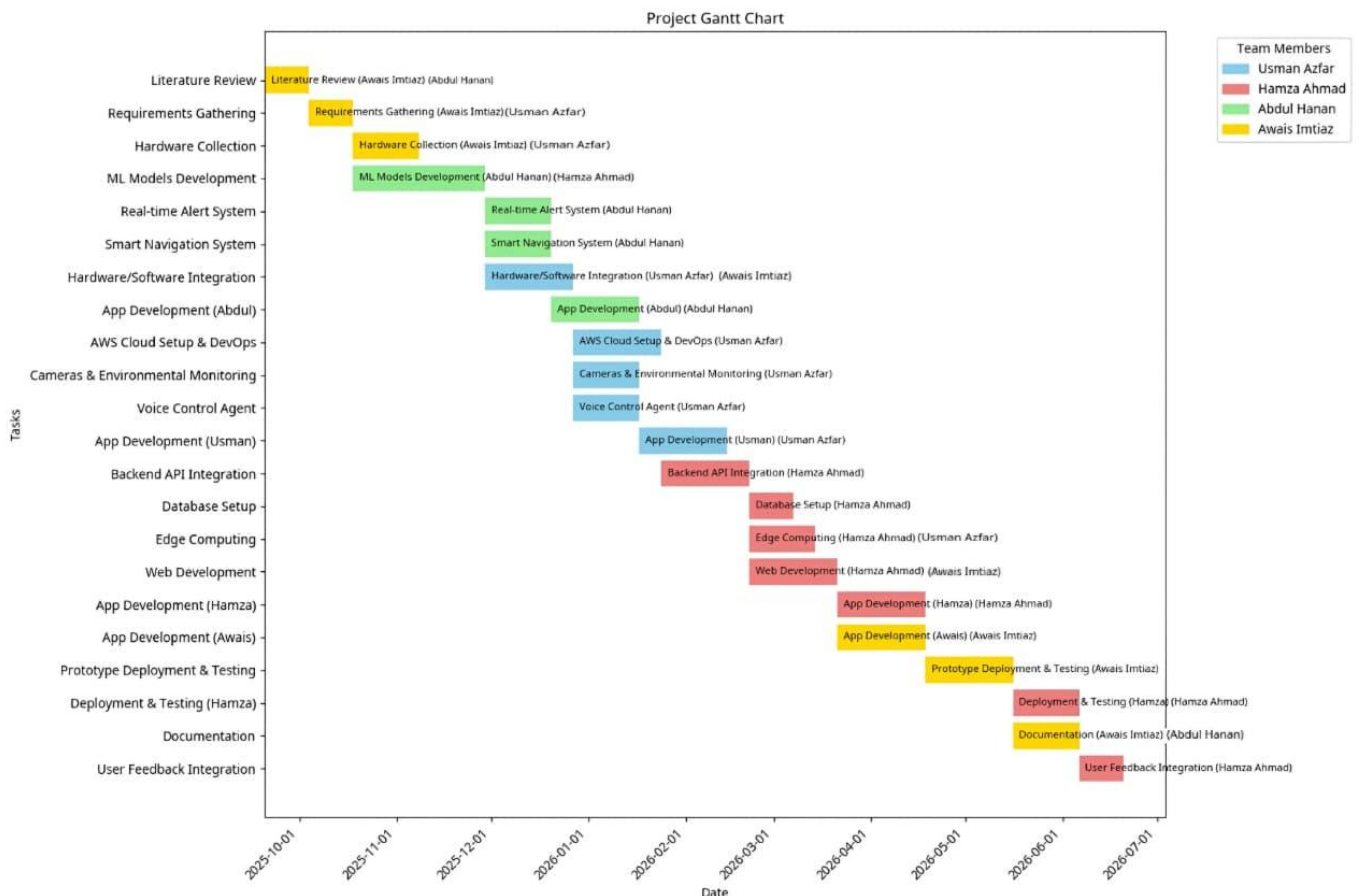


Fig 5.5 Full Development Life Cycle

6. Individual Tasks

Team Member	Activity	Tentative Date
Usman Azfar	Hardware Collection and Requirement Analysis, Hardware and software integration, AWS cloud setup and DevOps, Safety System, Voice Control Agent and Environmental Monitoring System, App Development	7-8 Weeks
Hamza Ahmad	Web development, app development, backend API integration, deployment and testing, edge computing, database, ML models testing, Hardware and Software integration, user feedback	8-9 Weeks
Abdul Hanan	Literature Review, ML models for crash and drowsiness detection, real time alert and communication system, smart navigation system, app development, Documentation	8-9 Weeks
Awais Imtiaz	Literature review, requirements, hardware collection and integration, prototype deployment and testing, web development, app development, API integration and testing, documentation.	7-8 Weeks

7. Gantt Chart



8. Tools and Technologies

- **Programming:** Python (ML), Kotlin (Android), JavaScript (Entire MERN stack for web).
- **ML:** TensorFlow Lite, OpenCV for vision, spaCy for voice NLP, Keras.
- **Cloud & Database:** AWS IoT Core, MongoDB, Twilio API
- **Hardware:** Appropriate cameras, sensors and edge computing according to market(optional).
- **Tools:** Android Studio, VS Code, Git for version control, Figma, Docker, AWS.
- **Others:** Google Maps API, Turnitin for plagiarism.

9. References

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- [2] S. Jaiswal, P. R. Jadhav, and A. R. Kale, “IoT based Vehicle Accident Detection and Alerting System,” *International Research Journal of Engineering and Technology (IRJET)*, vol. 2, no. 4, pp. 255–258, 2015. [Online]. Available: <https://www.academia.edu/download/37757852/Irjet-v2i204.pdf>
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- [5] R. Smith and P. Jones, “Heads-Up Displays in Automotive Applications,” *IEEE Transactions on Vehicular Technology*, vol. 69, no. 5, pp. 1234–1245, May 2020.
- [6] A. Sharma and R. Singh, “A Smart Helmet for Accident Detection Using IoT and Machine Learning and its Application in Applied Sciences,” *Applied Sciences*, vol. 11, no. 11, pp. 5039–5056, 2021. [Online]. Available: <https://www.mdpi.com/2076-3417/11/11/5039>

10. Appendix A

HelmX

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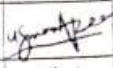
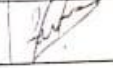
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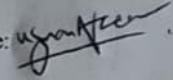
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Date: September 09, 2025

Name of Group Leader: Usman Azfar

Signature:



Name of Supervisor: Ms. Tayyaba Tariq

Co-Supervisor (if any):

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